

Doctor Appointment Booking System (Prescripto)

PHASE-I PROJECT REPORT

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Abstract

The Doctor Appointment Booking System, titled *Prescripto*, is a web-based application developed using the MERN stack — MongoDB, Express.js, React.js, and Node.js. The main aim of this project is to simplify and digitalize the traditional method of booking doctor appointments by creating a single online platform for patients, doctors, and administrators. Patients can easily register, search for doctors by specialization, and book appointments, while doctors can manage their schedules, update their availability, and monitor their appointments. The admin manages the entire platform by verifying doctors and monitoring all activities. The system includes secure authentication using JSON Web Tokens (JWT), password encryption with Bcrypt, and online payment integration through Razorpay and Stripe. Images and documents are managed using Cloudinary, ensuring smooth media storage. The frontend is designed using React.js with Tailwind CSS for a responsive and user-friendly interface. MongoDB is used to store all data efficiently. This system successfully automates the medical appointment process, reduces manual effort, and improves accessibility and convenience for both patients and doctors.

Keywords

Doctor appointment, MERN, Telemedicine, JWT Authentication, Online Payments

Introduction / Background

In today's busy world, scheduling a doctor's appointment through traditional methods like phone calls or in-person visits can be time-consuming and inefficient. The increasing use of technology in healthcare has created a need for an online platform that allows patients to easily connect with doctors. The *Doctor Appointment Booking System* aims to simplify this process by providing a digital solution where patients can register, browse doctors based on specialization, and book appointments instantly. Doctors can manage their availability, view patient requests, and handle appointments conveniently through their own dashboards. The system also provides an admin module to supervise and manage users, doctors, and appointments securely. With integrated payment gateways and cloud-based data storage, the platform ensures smooth transactions and reliable data access. Overall, this project enhances healthcare accessibility, saves time, and promotes efficiency for both patients and doctors.

Problem Statement

In most hospitals and clinics, patients still depend on manual methods or phone calls to book appointments. This often leads to confusion, overlapping schedules, and long waiting times. Doctors also find it difficult to manage their availability and patient lists efficiently. To overcome these issues, there is a need for an online system that allows patients to book appointments easily, helps doctors manage their schedules, and provides a clear and organized platform for both sides.

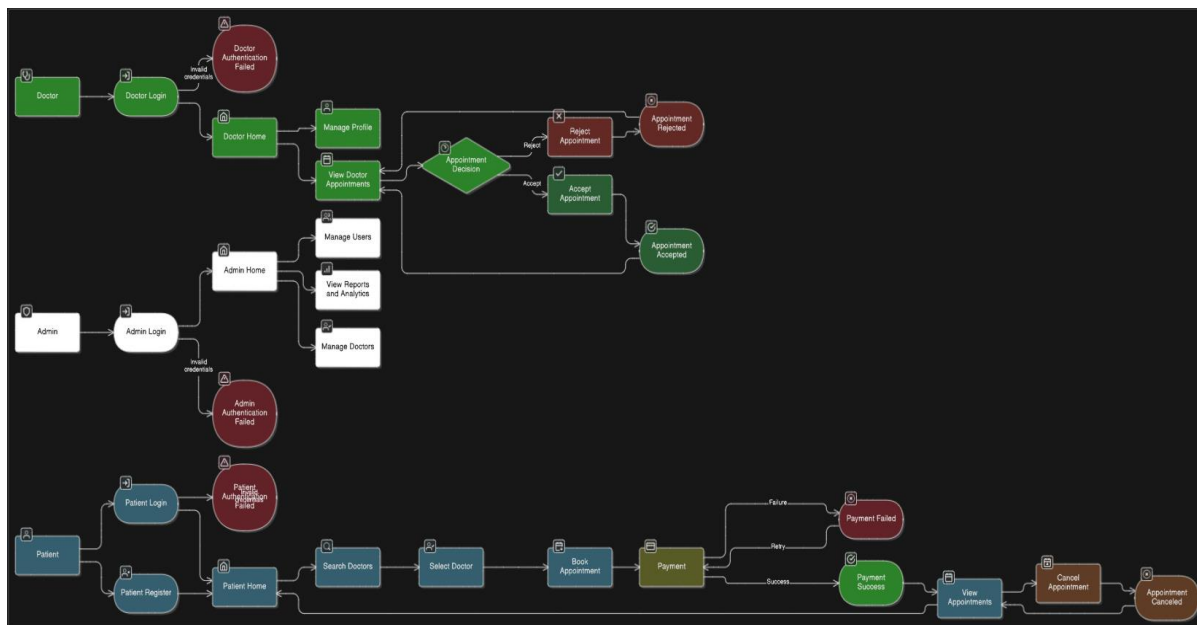
Objectives

- To create a web-based platform for online doctor appointment booking.
- To allow patients to register, log in, and book appointments in a simple way.
- To provide doctors with tools to update their availability and view appointments.
- To add a secure online payment feature for appointment fees.
- To include admin control for managing users and maintaining the system properly.

Scope & Limitations

- The system includes three roles — Patient, Doctor, and Admin.
- Patients can search for doctors, book appointments, and make payments.
- Doctors can view, accept, or cancel appointment requests.
- Admin can manage both users and doctors from a single dashboard.
- The project works only with an internet connection.
- It focuses on appointment booking and does not include full medical record management

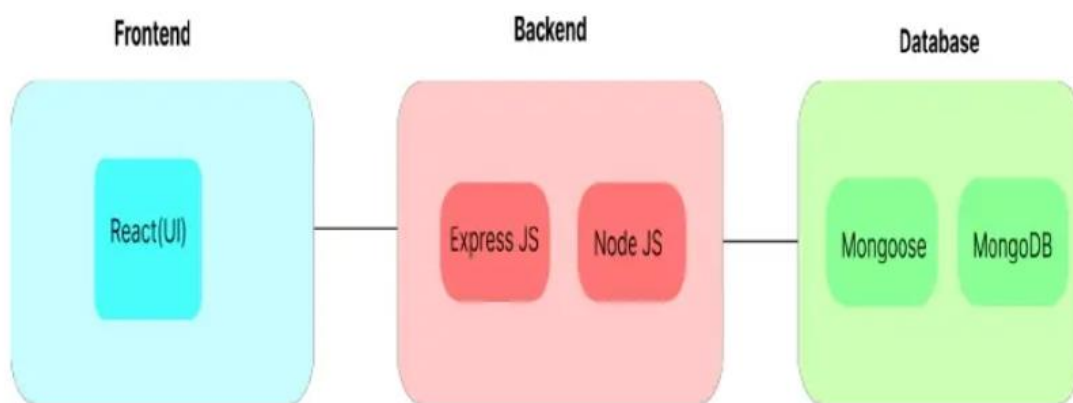
Use Case Diagram



Technology Stack

- Frontend: React.js (Vite) with Tailwind CSS
- Backend: Node.js, Express.js
- Database: MongoDB (Mongoose) / MongoDB Atlas
- Authentication: JWT, bcrypt
- Payments: Razorpay and Stripe (server-side integrations)
- Media: Cloudinary for image uploads
- Realtime: Agora SDK for video (optional)
- Deployment: Vercel (frontend), Render/Heroku/VPS (backend)

System Architecture



In this architecture diagram:

1. The frontend is represented by the "Frontend" section, including user interface components such as User Authentication, application, Profile, Admin dashboard, etc.,
2. The backend is represented by the "Backend" section, consisting of API endpoints for Users, application, conformation, etc., It also includes Admin Authentication and an Admin Dashboard.
3. The Database section represents the database that stores collections for Users, Admin, application, and conformation

Data Model / Key Collections

The database is designed using a **document-based model** (MongoDB), which allows flexible data storage for patients, doctors, and appointments.

Entity	Key Fields	Description
User	userId, name, email, password, role	Stores login and profile details for patients and doctors.
Doctor	doctorId, name, specialization, experience, fees, image	Contains professional details of doctors available for appointments.
Appointment	appointmentId, userId, doctorId, date, time, status, payment	Keeps records of all bookings and their statuses.
Admin	adminId, email, password	Manages and monitors the overall system.

Inputs:

- User registration details
- Doctor information and schedules
- Appointment booking details

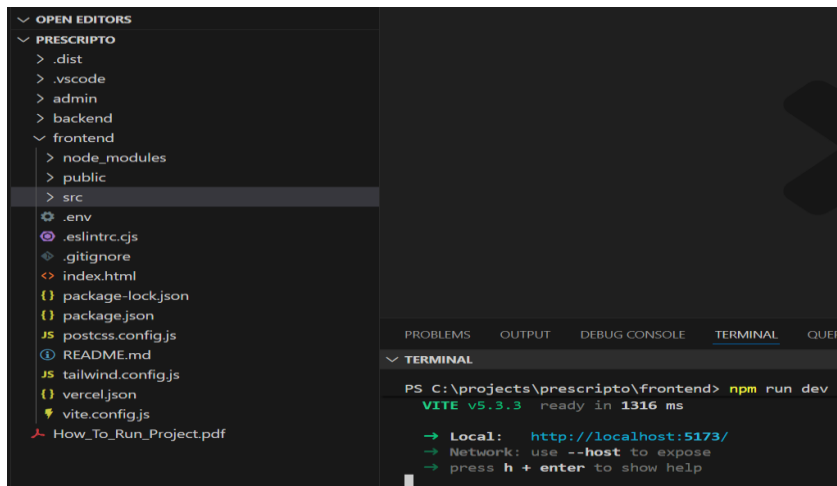
Outputs:

- Appointment confirmations
- Notifications and receipts
- Reports for admin overview

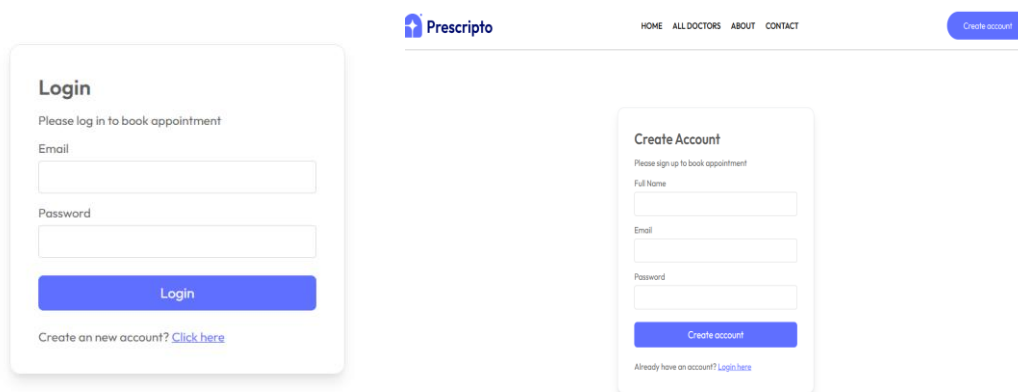
Key MongoDB collections and important fields:

1. users: { _id, name, email, password(hash), role, phone, address, image }
2. doctors: { _id, name, speciality, fees, slots_booked, available, profile, image }
3. appointments: { _id, userId, docId, slotDate, slotTime, amount, payment(boolean), cancelled(boolean), date }
4. payments: { _id, appointmentId, provider, amount, status, transactionId, date }

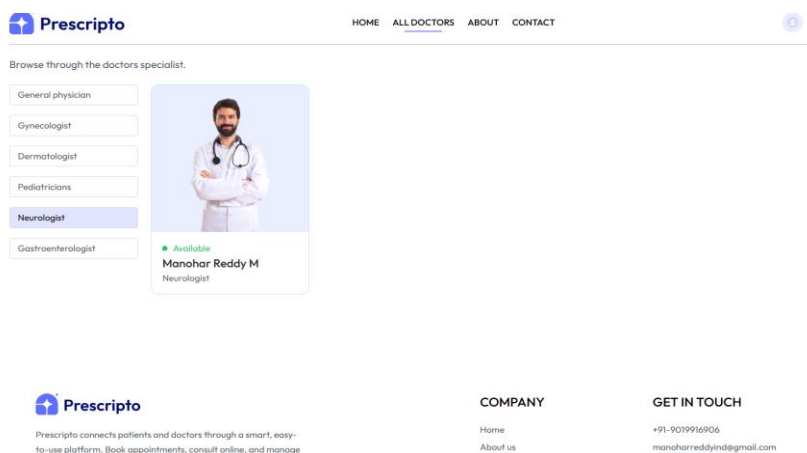
Step-by-Step Worked Procedure



The frontend of the project is developed using React and Tailwind CSS. The app runs using the command `npm run dev` and is hosted locally on port 5173.

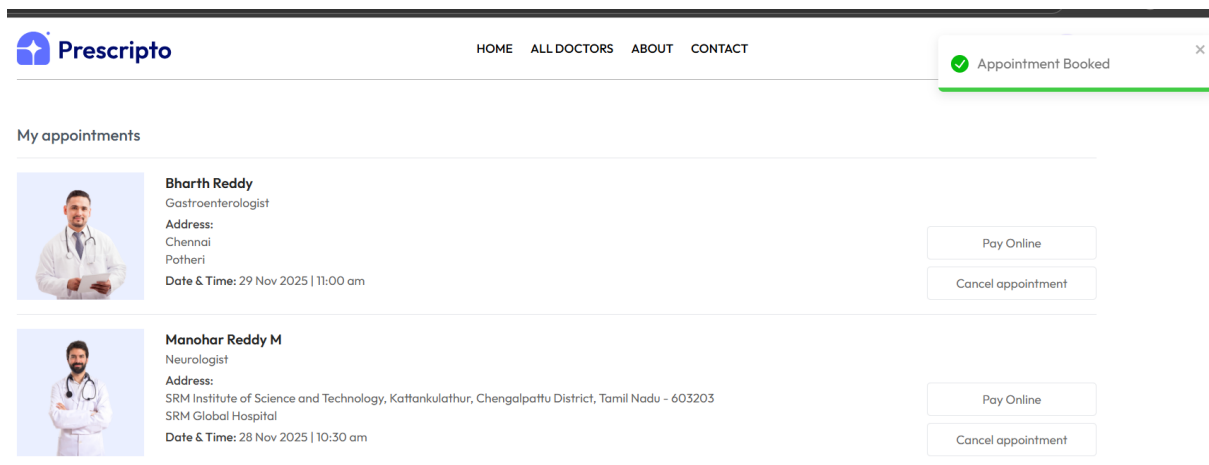


This is the login interface for patients and doctors. After successful authentication using JWT, users are redirected to their dashboard.

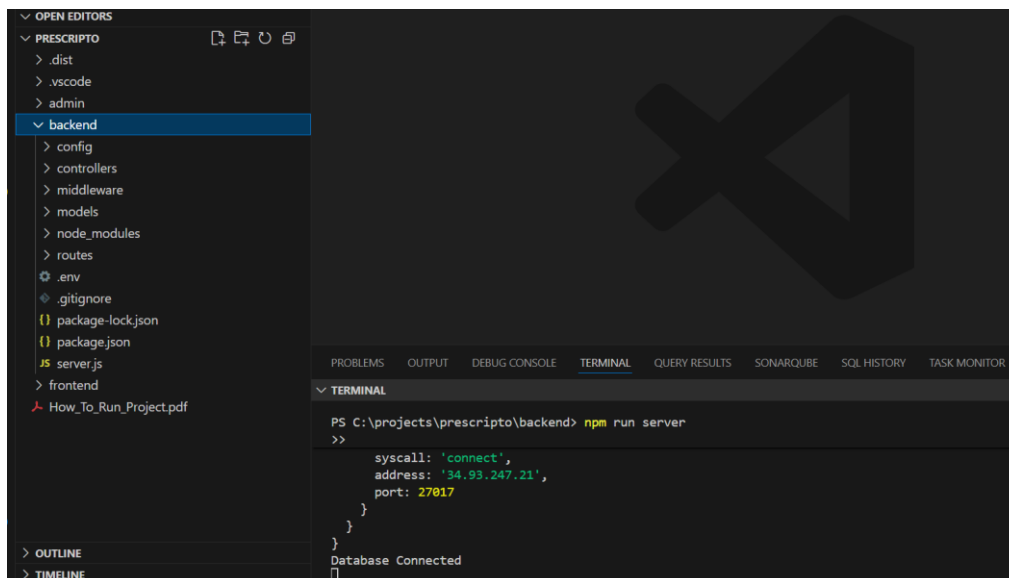


Patients can search for doctors based on specialization, availability, or location.

The data is fetched from the backend through an API request.



Once a doctor is selected, the patient confirms the date and time. The appointment details are stored in MongoDB and a payment option is displayed.



The backend is built using Node.js and Express.

It handles user authentication, doctor management, and appointment APIs.

The Doctor Appointment Booking System successfully simplifies the process of scheduling medical consultations by connecting patients, doctors, and administrators on one digital platform. It allows patients to book appointments, make payments, and communicate with doctors conveniently from anywhere. For doctors, it provides an easy way to manage schedules, view patient details, and track earnings, while administrators can oversee all users and activities efficiently. The project not only improves accessibility but also ensures transparency and security through proper authentication and data handling. Overall, this system represents a step toward modernizing healthcare management using web technologies and demonstrates the practical application of the MERN stack in real-world problem-solving.